Biliary tract cancer

Gallbladder · intrahepatic, perihilar and distal cholangiocarcinoma · **shared systemic therapy; surgery and molecular profile differ by site**

Compiled from the Stanford Oncology Handbook 2025, Gallbladder Cancer and Cholangiocarcinoma (Northness A, Heestand G) · NCCN · primary trial reports and FDA actions verified against PubMed, September 2026.

FOUR DISEASES IN ONE NAME

Site	Notes
Intrahepatic CCA *~10% of CCA*	Arises proximal to the second-order ducts. **The molecularly rich one** — FGFR2 and IDH1 live here
Perihilar (Klatskin) *~40% of CCA*	At the confluence of the right and left hepatic ducts. Resection planned by **Bismuth-Corlette class**
Distal CCA *~50% of CCA*	Common bile duct up to, but not including, the ampulla
Gallbladder cancer	<5,000 cases/yr US; fundus 60% > body 30% > neck 10%. **1–2% are found incidentally** at cholecystectomy for stones

~8,000 cholangiocarcinomas/yr in the US, median age ~70. **The subtypes share systemic therapy but differ in staging system, operation, and molecular profile** — which is why anatomic site has to be stated before anything else. **Risk:** CCA — PSC, HBV/HCV, liver flukes, male sex, obesity, diabetes, alcohol. **GBC** — cholelithiasis, gallbladder calcification, female sex, chronic Salmonella or H. pylori infection.

WORKUP

- **Labs:** CBC with differential, CMP, PT/INR, **CA 19-9**, HBV and HCV serologies
- **Imaging:** triphasic CT abdomen/pelvis or **liver-protocol MRI** + CT chest to complete staging. **PET/CT** to exclude occult disease when the plan is resection
- **Gallbladder:** start with ultrasound — go to triphasic CT or MRI for a **polyp >10 mm** or wall thickening not explained by cholecystitis
- **MRCP** to plan biopsy and stenting. **ERCP** to stent, decompress and brush — **brushings are often non-diagnostic, so go to EUS-FNA** when they are
- **Genomically profile every tumor**, and send **MMR, HER2 IHC and PD-L1**. This is not optional in biliary cancer — roughly a third of patients have a level 1/2 actionable alteration
- **If perihilar CCA might go to transplant, do NOT perform a transperitoneal biopsy** — tumor seeding disqualifies the patient from the protocol

TREATMENT PATHWAY

Resectable

Resection by anatomic site
gallbladder T1a: cholecystectomy alone
everything else: formal resection + lymphadenectomy

Adjuvant therapy
capecitabine ×8 cycles (BILCAP) is the default
node-positive or margin-positive derive the clearest benefit — consider adding chemoradiation (SWOG S0809)

Surveillance
exam, **CA 19-9** and triphasic CT C/A/P
q3 months ×2 years, then q6 months

Locally advanced, unresectable

Systemic therapy first
treat as metastatic — gem/cis + durvalumab or pembrolizumab

Then consider local therapy
HAI floxuridine, Y-90 or ablative RT
at experienced centers — **none randomized-proven**

Metastatic

Start chemo-IO and profile in parallel
gem/cis + durvalumab (TOPAZ-1) **or + pembrolizumab** (KEYNOTE-966) — near-identical benefit, no biomarker
send NGS, MMR and HER2 IHC on day one

Second line
targeted therapy if an alteration was found — see page 2
otherwise **FOLFOX** (ABC-06)

STAGING — THREE SEPARATE AJCC SYSTEMS

Site	T staging hinges on
Intrahepatic	**Size and vascular invasion:** T1a ≤5 cm / T1b >5 cm, both without vascular invasion · T2 vascular invasion *or* multiple tumors · T3 perforates visceral peritoneum · T4 direct extrahepatic invasion. **N is simply N0 / N1**
Perihilar	**Depth then vessels:** T1 confined to the duct · T2a into surrounding fat, T2b into hepatic parenchyma · T3 unilateral portal vein or hepatic artery branch · T4 main portal vein bilaterally, common hepatic artery, or second-order radicals with contralateral vessel
Distal	**Millimetres of duct-wall invasion:** T1 <5 mm · T2 5–12 mm · T3 >12 mm · T4 celiac axis, SMA or common hepatic artery
Gallbladder	**Which side of the wall:** T1a lamina propria, T1b muscularis · **T2a peritoneal side, T2b hepatic side** — the distinction drives the operation · T3 serosa, liver or one adjacent organ · T4 main portal vein / hepatic artery or ≥2 organs

Nodes: perihilar, distal and gallbladder all use **N1 = 1–3 nodes, N2 = ≥4**. Intrahepatic uses N0/N1 only.

SURGERY

- Most patients are unresectable at presentation, and **most resected patients still recur**
- **Resectability requires:** no retropancreatic or paraceliac nodes, no distant liver metastases, no main portal vein or hepatic artery invasion, no adjacent-organ invasion, no dissemination
- **Intrahepatic** → hepatic lobectomy; 5-yr OS 11–40%. **Perihilar** → en bloc resection by Bismuth-Corlette class. **Distal** → Whipple + extrahepatic duct; 5-yr OS 10–40%
- **Gallbladder: T1a is cured by simple cholecystectomy alone. T1b and above need en bloc cholecystectomy + resection of liver segments 4a/4b + regional lymphadenectomy**
- **Incidentally discovered gallbladder cancer beyond T1a must go back to the operating room** for the 4a/4b bed resection and lymphadenectomy
- **Liver transplant** is an option for highly selected **perihilar** CCA (Mayo protocol: mass ≤3 cm, no transperitoneal biopsy, no metastases) — intention-to-treat 5-yr survival ~53%

Biliary tract cancer

ALTERATIONS BY SITE

	iCCA	eCCA	GBC
FGFR2 fusion / rearrangement	**~12%**	~1%	~1–2%
IDH1	**~20%**	~1%	~1%
IDH2	~6%	0%	~0.5%
HER2 / ERBB2	~4%	**~9%**	**~11–14%**
BRAF V600E	~3%	~0%	~0%
KRAS, any	~12%	**~38%**	~10%
MSI-H / dMMR	~1.5%	~4%	~2%
BAP1	**~19%**	~2%	~1%
TP53	~21%	~44%	**~60%**

FGFR2 and IDH1 are intrahepatic diseases and are mutually exclusive. HER2 is a gallbladder and extrahepatic disease. Roughly **a third of all biliary cancers** carry a level 1/2 actionable alteration — but that is **~40% of intrahepatic** versus **~15% of extrahepatic**. Western cohorts (AACR GENIE); in East Asian series IDH1 and FGFR2 fall and KRAS rises.

LIVING WITH AN FGFR INHIBITOR

- **Hyperphosphatemia is on-target**, not idiosyncratic — FGFR1 blockade removes the FGF23 phosphaturic signal. It is a marker that the drug is working. **Pemigatinib ~60%, futibatinib ~85%, lirafugratinib ~21%**. Manage with a low-phosphate diet, **sevelamer**, and dose holds
- **Ocular toxicity:** serous retinopathy / retinal pigment epithelial detachment, plus dry eye in over 20%. **Baseline and serial ophthalmology with OCT** — roughly q2 months for 6 months, then q3 months
- **Nail and hair:** onycholysis in ~47%, alopecia in a third. Stomatitis, arthralgia and diarrhea are common
- **Resistance arrives at ~6–9 months** as **polyclonal secondary FGFR2 kinase-domain mutations**, classically the gatekeeper **V564F**. The covalent agents (futibatinib, lirafugratinib) retain activity against some but not all of these

DRIVER → THERAPY

Alteration	Therapy
FGFR2 fusion or rearrangement *~12% of iCCA*	**All second-line and beyond. Lirafugratinib** (FGFR2-selective; **approved 23 Sep 2026**) — ORR 46%, DOR 11.8 mo, and notably **less hyperphosphatemia (~21%)**. **Futibatinib** ORR 42% (accelerated; **label is intrahepatic only**). **Pemigatinib** ORR 37% (accelerated, any site). **Only fusions and rearrangements qualify — point mutations do not.** FIGHT-302 moved pemigatinib to 1L on PFS but **showed no OS gain**, and the 1L indication is not approved. **Infigratinib was withdrawn** in 2024
IDH1 *~20% of iCCA*	**Ivosidenib** after 1–2 prior lines (regular approval). **PFS 2.7 vs 1.4 months, HR 0.37** — statistically emphatic, clinically modest. Unadjusted OS was not significant (10.3 vs 7.5 mo) because **70% crossed over**; crossover-adjusted HR 0.49
HER2 *GBC ~11–14%, eCCA ~9%*	**Zanidatamab** — the only BTC-specific approval (accelerated, Nov 2024), **IHC 3+ only**, 2L+: ORR **52%** in IHC 3+, DOR 14.9 mo. **T-DXd** is covered by the tumor-agnostic IHC 3+ approval, but **ILD reached 17% in the biliary cohort** — higher than in most tumor types. **Trastuzumab + pertuzumab is NCCN-listed but not FDA-approved here**
BRAF V600E *~3% of iCCA*	**Dabrafenib + trametinib** — tumor-agnostic accelerated approval. ROAR biliary cohort ORR 47–53%, PFS 9.0 mo
MSI-H / dMMR *~1–3%*	**Pembrolizumab** (tumor-agnostic, now regular approval) — KEYNOTE-158 cholangiocarcinoma cohort **ORR 41%**, median OS 24.3 months. Dostarlimab is an alternative
TMB ≥10 mut/Mb *~4%*	Pembrolizumab, tumor-agnostic accelerated approval
NTRK or RET fusion *<1% each*	Larotrectinib / entrectinib / repotrectinib for NTRK; **selpercatinib** for RET (tumor-agnostic, converted to traditional approval in 2026). **Pralsetinib has no tumor-agnostic approval**
KRAS G12C *~1%*	**No approved option in biliary cancer.** Adagrasib showed activity in a small pan-tumor cohort (5/12); trial or off-label only

ADJUVANT THERAPY

- **Capecitabine ×6 months (BILCAP) is the standard**; **S-1** (ASCOT) where available (Japan; not in the US). **Add chemoradiation (SWOG S0809) for node-positive or margin-positive** extrahepatic CCA or gallbladder cancer

LIVER-DIRECTED THERAPY

- **HAI floxuridine, Y-90 and ablative RT/SBRT** are options at experienced centers for liver-dominant disease — **none is randomized-proven**

REGIMEN DOSING

Regimen	Dosing
Gem/cis + durvalumab *TOPAZ-1, 1L*	Gemcitabine **1000 mg/m²** + cisplatin **25 mg/m²** on **days 1 and 8**, with **durvalumab 1500 mg** day 1 — **q21d**. Chemotherapy for up to 8 cycles, durvalumab continues
Gem/cis + pembrolizumab *KEYNOTE-966, 1L*	Same gem/cis backbone with **pembrolizumab 200 mg** day 1, q21d
FOLFOX *ABC-06, 2L*	Oxaliplatin **85 mg/m²** day 1 + 5-FU **1200 mg/m²/day ×48 h** continuous infusion from day 1 — **q14d**
Gem + nab-paclitaxel	Nab-paclitaxel **125 mg/m²** then gemcitabine **1000 mg/m²**, **days 1, 8, 15 of a 28-day cycle**. An alternative, not a standard
Adjuvant capecitabine *BILCAP*	**1250 mg/m² PO twice daily, days 1–14 of a 21-day cycle, ×8 cycles** (6 months). Grade 3 hand-foot in ~20%
Adjuvant chemoRT *SWOG S0809*	Gemcitabine + capecitabine ×4 cycles, then **capecitabine with radiation ×6 weeks** (45 Gy nodal, 54–59.4 Gy tumor bed)
Ivosidenib	**500 mg PO daily** — IDH1-mutant, after 1–2 prior lines
Lirafugratinib	**70 mg PO daily** — FGFR2 fusion/rearrangement, previously treated, FGFR-inhibitor-naive

KEY TRIALS

ABC-02	Gemcitabine + cisplatin vs gemcitabine — OS **11.7 vs 8.1 mo**, HR 0.64. The backbone, since 2010
TOPAZ-1	Add **durvalumab** to gem/cis — OS 12.8 vs 11.5 mo, HR 0.80; **24-mo OS 24.9% vs 10.4%**
KEYNOTE-966	Add **pembrolizumab** to gem/cis — OS 12.7 vs 10.9 mo, HR 0.83. Effectively interchangeable with TOPAZ-1
ABC-06	**FOLFOX** after gem/cis — OS 6.2 vs 5.3 mo; **12-mo OS 25.9% vs 11.4%**
SWOG S1815	**Do not add nab-paclitaxel** to gem/cis — no OS gain
BILCAP	Adjuvant capecitabine — OS 51.1 vs 36.4 mo; **standard of care**
ASCOT	Adjuvant **S-1**, Japan — 3-yr OS 77.1% vs 67.6%. Not applicable in the US
FIGHT-202 / FOENIX-CCA2	Pemigatinib (ORR 37%) and futibatinib (ORR 42%) in FGFR2-altered disease
ReFocus	**Lirafugratinib** — ORR 46%, DOR 11.8 mo, far less hyperphosphatemia. **Approved 23 Sep 2026**
ClarIDHy	Ivosidenib in IDH1-mutant — PFS 2.7 vs 1.4 mo, HR 0.37
HERIZON-BTC-01	Zanidatamab in HER2 IHC 3+ — ORR 52%, DOR 14.9 mo
ROAR	Dabrafenib + trametinib in BRAF V600E — ORR 47–53%